Using Machine Learning for Analyzing Yelp Reviews

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Abstract

The primary goal of this paper was to investigate how a simple machine learning model, trained on pre-processed text data, performs in classifying the sentiment of Yelp reviews. The model was trained using LASSO regularization in a logistic regression of sentiment (whether a review was above or below three stars) on the counts of various words, which had the benefit of selecting the words that were most predictive of review sentiment. A secondary goal was to employ another LASSO model to detect which words drive engagement in reviews, running a post-LASSO model to assess statistical significance. The results suggest that a LASSO regularized model does well in selecting words with positive or negative connotations when predicting sentiment; however, more work needs to be done to determine what drives review engagement.

1 Introduction

Online review platforms like Yelp, which allows users to rate a variety of businesses on a scale of one to five stars, allow prospective customers to quickly gauge whether they would like to visit a given business. Thus, with the rise of the Internet, online ranking systems such as Yelp have become increasingly important in determining which businesses can attract new customers. Due to their widespread use, online review platforms provide an opportunity to study consumer sentiment on a large scale. That is, consumers using these platforms directly provide a measure of their sentiment (stars, thumbs up/thumbs down, etc.) and brief descriptions corresponding to that sentiment, which in turn makes it possible to use this information to train machine learning models for quickly classifying reviews based on the language they contain. The reason this might be useful is that not all review platforms use the same measures of sentiment, which might complicate analysis across different platforms; thus, training a model to identify whether the string of text in a review is positive or negative could potentially be generalized to extract sentiment from any string of text, even those on non-review platforms. Using data on 10,000 Yelp reviews from 2007-2013, my primary goal in this paper was to train a logistic regression using LASSO regularization to classify reviews' sentiment (whether they were positive or negative) using counts of the words contained in their descriptions. My approach using word counts to extract sentiment from Yelp reviews was inspired by Gentzkow and Shapiro (2010), who used counts of phrases in the congressional record to construct an index of newspaper slant.

Another aspect of online ranking systems that might be worth studying is the potential for businesses to manipulate such systems in their favor (Goldfarb and Tucker 2019). If having more positive reviews on average is associated with businesses receiving more customers and therefore higher profits, this may provide an incentive for them to create fictitious reviews aimed at inflating their overall ranking. Naturally, businesses engaging in this strategic behavior may be interested in not only constructing positive reviews, but reviews that are the most likely to grab the attention of prospective customers. A convenient feature of Yelp is that it allows users to react to other people's reviews, which can be used to construct a measure of total engagement for a given review. Thus, a secondary goal in this paper was to train a simple LASSO model to predict total review engagement based on the language used in reviews' descriptions and use a post-LASSO method to investigate the degree to which specific words were the most predictive of engagement.

2 Data

The dataset used for my analysis was freely available on kaggle.com and contained information on 10,000 Yelp reviews from between 2007 and 2013 (Sabnis 2018). For each review, it contained ID variables for the reviewers and businesses, the dates the reviews were posted, the ranking given by the user on a scale of one to five stars, and measures of engagement in the form of counts of "funny", "cool", and "useful" reactions. To construct a measure of review sentiment for use in a logistic regression for classifying reviews, I defined review sentiment as an indicator variable for whether a reviewer assigned 3 or more stars to a given business:

$$sentiment_i = 1(stars_i \geq 3),$$

where a value of 1 corresponds to a positive review and a value of 0 corresponds to a negative review. The main assumption in constructing this indicator variable is that reviewers with a positive opinion of a given business will always assign the majority of stars available (at least three out of five), whereas reviewers with negative opinions of a business will assign one or two stars. This assumption may not hold for reviewers with negative opinions who might assign three stars to avoid coming off as being harsh.

Lastly, I constructed a measure of review engagement as simply the sum of the counts of the different types of reactions that users could assign each other's reviews:

$$engagement_i = cool_i + funny_i + useful_i,$$

where $cool_i$ is the count of "cool" reactions for a given review, $funny_i$ is the count of "funny" reactions, and $useful_i$ is the count of "useful" reactions. The main assumption in taking the sum of all reactions rather than studying what drives each type of reaction individually is that businesses engaging in strategic behavior may not be interested how exactly people are reacting to their fictitious reviews; rather, they might only care about maximizing the amount of people that see their reviews.

3 Methods

My analysis in this paper consisted of three main steps. First, in order to convert the raw text from Yelp review descriptions into variables that are usable in standard regression and supervised learning models, I had to employ a variety of text pre-processing techniques to extract counts of unique words from each review. Next, I used counts of the 1,000 most reoccurring words across all reviews in the sample to train a model of review sentiment using a logistic regression with LASSO regularization, and investigated how well such a model performs in classifying whether a given review's sentiment was positive or negative. Lastly, I trained another LASSO model to identify which words are the most predictive of engagement within a positive review, employing a post-LASSO procedure to assess statistical significance.

3.1 Text pre-processing

Since raw text alone is is typically unusable when included as a variable in conventional regression specifications, I used a variety of methods to extract counts of the unique words from each Yelp review in order to assign each word a numerical value. The steps for pre-processing the raw text from Yelp reviews, in no particular order, were the following:

1. Converting all text to lower case: By doing so, words are no longer case sensitive and therefore will add to the same total count regardless of where they appear in a sentence.

2. Removing punctuation: Since I was only interested in training models using counts of words, punctuation was not relevant to my analysis.

3. Removing whitespace: Removing discrepancies in spacing (tabs, new lines, extra spaces) reduces the possibility that spaces will be included in the final output, which may result in identical words being counted differently.

4. Removing stop words: Words such as "and", "or", and "of", were excluded to simplify the analysis, since such words are not opinionated and likely offer little to no information about the sentiment of a Yelp review.

5. Lemmatizing words: Lemmatization is the process of stripping different conjugations of words down to their roots to count them all as occurrences

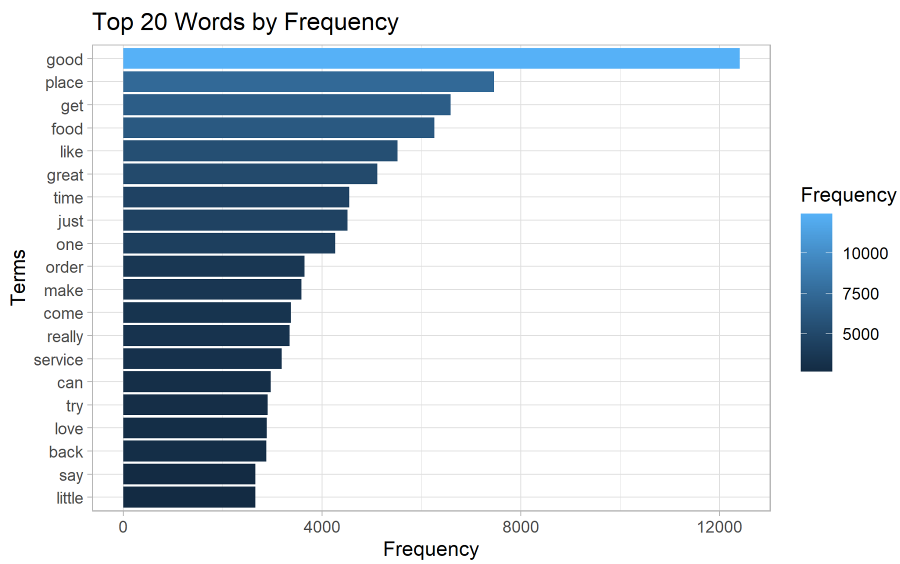

Figure 1: Top 20 words by frequency

of the same word. For example, using lemmatization, a series of words like "run", "ran", and "running" would be counted as three occurrences of the word "run".

Additional steps could include removing numerical characters from each Yelp review, however I did not find that to be necessary for the sake of my analysis. Lastly, after following the five steps outlined above, I extracted each unique word in the sample into a term document matrix containing counts of each word in all 10,000 reviews. The resulting matrix contained counts of 29,199 unique terms; however, to reduce the computational load of handling this many variables, I merged counts of only the 1,000 most frequently occurring words with the original dataset.

Figure 1 displays the 20 most frequent words across all 10,000 reviews following the text pre-processing stage. The dataset did not contain information on what types of businesses were being reviewed, but the word "food" being the fourth most common word in the dataset suggests that many of the reviews were for restaurants or other food-related establishments. All of the words included in this figure, plus 980 additional words, were added to the original dataset.

3.2 Training a model of review sentiment

To train a logistic regression of review sentiment on the counts of the most frequently occurring words in the dataset using LASSO regularization, I employed a 70/30 train/test split, training the model on 70% of the original dataset using a 3-fold cross-validation procedure and evaluating its performance on the test set. Due to the logistic regression specification, the resulting predicted values of review sentiment took values between 0 and 1, with 0 representing negative reviews and 1 representing positive reviews. The motivation behind using LASSO regularization was due to its ability to conduct model selection by setting the coefficients on words that might be less predictive of review sentiment to 0. As a result, the words selected to be included in the LASSO model may offer insight into which specific words are the most indicative of review sentiment.

In order to convert continuous fitted values to binary classifications, I evaluated the model's predictive accuracy out of sample for a range of cutoff rules, and chose the optimal cutoff as being that which maximized the positive difference between the true positive rate (the proportion of correctly classified positive reviews) and the false positive rate (the proportion of incorrectly classified positive reviews). This method is fairly easy to interpret, as this difference for an ideal classifier would be equal to 1, implying a 100% true positive rate and a 0% false positive rate (or a 100% true negative rate). Therefore, the the closer this difference is to 1, the more accurate the classifier is.

3.3 Training a model of review engagement

To train a model of review engagement, I used a similar approach to what I used for the model of review sentiment, regressing review engagement on counts of the most frequently occurring words in the dataset using LASSO regularization. For this approach, I used a 70/30 train/test split for a subset of positive reviews (n = 8,324) and trained the model using a 3-fold cross-validation procedure, evaluating its performance on the test set. Finally, I employed a post-LASSO procedure, running an OLS regression using all 8,324 observations in the subset of positive reviews to assess the statistical significance of the words chosen to be the most predictive of review engagement.

The reason I used only a subset of positive reviews was that my goal in estimating this model was to inspect which words would be of interest to businesses manipulating online ranking systems in their favor. Naturally, these businesses might be more likely to leave positive reviews on their own pages, so studying what drives engagement in positive reviews might offer insight into how businesses engaging in strategic behavior might try to draw the attention of potential customers. A key assumption here is that businesses would only be motivated by engagement with positive reviews.

4 Results

4.1 Model of review sentiment

Table 1 shows the variables with the largest coefficients in terms of magnitude in the model of review sentiment described in section 3.2. The results suggest that LASSO regularization did very well in identifying opinionated words, as the top words with positive and negative coefficients in the LASSO model are unsurprising for a model trained on review data, which is inherently opinionated. Some exceptions include words like ”haven’t” and ”spin.dry”, whose inclusion in the model may be due to spurious correlations in the data. Furthermore, the inclusion of ”spin.dry” as a regressor suggests that the steps followed in the pre-processing stage did not fully remove punctuation in the dataset; however, this may not be a concern considering that the other variables selected by the LASSO model have obvious associations with the outcome of interest.

Table 1: Top Words by the Magnitude of Their Coefficients

Words with Positive Coefficients	Words with Negative Coefficients
excellent	awful
fantastic	poor
professional	gross
perfect	bland
satisfy	horrible
beat	overpriced
delicious	terrible
amaze	mediocre
havent	rude
awesome	obviously
love	mess
favorite	suppose
authentic	sorry
great	suck
super	spin.dry
reasonably	sad
share	somewhere
juicy	waste
wonderful	bad
mood	dirty

Figure 2 illustrates the out of sample performance of the model of review sentiment in classifying whether a given review was positive or negative for a range of possible classification rules, or thresholds for determining whether predicted values from the LASSO-regularized logistic regression should be converted to zeroes or ones. The performance of the model was evaluated using a receiver operating characteristic (ROC) curve, which illustrates the tradeoff between a classifier’s true positive rate and its false positive rate. The black point in the graph represents the true positive rate and false positive rate for the model of review sentiment with a classification rule of 0.85, which was chosen by maximizing the difference between the two rates across many possible classification rules. For comparison, I also plotted the ROC curve for a random classifier with predicted values from a uniform distribution between 0 and 1 and a classification rule of 0.5.

Overall, the LASSO-regularized logistic regression model of review sentiment

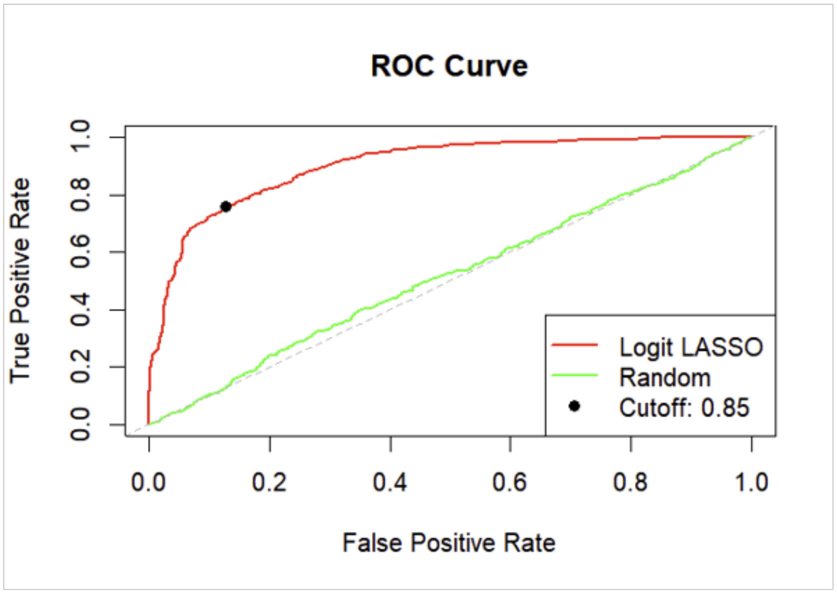

Figure 2: ROC Curves for review classification

does fairly well at classifying review sentiment on new data, as it has a true positive rate of 79.39% and a false positive rate of 16.02%. This illustrates that even a somewhat crude method of classifying strings of text, using only counts of individual words as variables, can do quite well in classifying unseen data. Thus, with a more robust model incorporating phrases of words and more complex measures of sentence structure, it is likely that the accuracy of classification could improve. However, for the purposes of this paper, I was only interested in evaluating how well a simple machine learning model can perform in detecting review sentiment. Nevertheless, my model does far better than a random classifier in detecting review sentiment, so there is undoubtedly some merit to using counts of individual words in such a model.

4.2 Model of review engagement

When training a LASSO regularized regression of total review engagement on counts of the 1,000 most frequent words in the dataset, the relationship between the words selected by LASSO and the outcome is much less clear than it was for the model of review sentiment. To investigate why this might be the case, I calculated the LASSO model's in and out of sample R^2 values, denoted as R^2_{IS} and R^2_{OOS}. The resulting values for R^2_{IS} and R^2_{OOS} were 16.41% and 10.23% respectively, suggesting that variation in the counts of words selected by the

LASSO model does not explain much of the variation in review engagement in or out of sample.

Although my model of review engagement performed poorly in predicting engagement, I wanted to get a sense of the degree to which the words chosen by LASSO regularization were predictive of engagement. Thus, I employed a post-LASSO procedure, running an OLS regression of engagement on counts of the words selected by LASSO for the entire subset of positive reviews. Table 2 displays the OLS regression results for the words with the 20 largest coefficients in terms of magnitude. While many of the words chosen by the LASSO model have statistically significant coefficients, not accounting for heteroskedasticity, the practical significance is still unclear for most of the words that were selected. The word "damn" having the largest coefficient in the model might suggest that more expressive reviews contribute more to their overall engagement, but it is hard to imagine that the inclusion of words like "cafe" or "garden" would have any effect on how people react to a given review. Because of its poor predictive performance and lack of logical interpretations on its coefficients, a model of review engagement trained on word counts alone is insufficient for studying how businesses might manipulate online ranking systems in their favor.

Table 2: OLS regression output for top 20 post-LASSO estimates by magnitude

	Estimate	Std. Error	t value	Pr(>|t|)
damn	3.214	0.446	7.211	0
lucky	2.804	0.544	5.157	0
hey	2.729	0.597	4.572	0
kick	2.182	0.444	4.917	0
blow	2.090	0.502	4.161	0
sorry	1.884	0.555	3.394	0.001
X100	1.838	0.483	3.804	0.0001
brunch	1.754	0.358	4.897	0
garden	1.712	0.437	3.922	0.0001
cafe	1.641	0.363	4.518	0
enter	1.603	0.497	3.225	0.001
patron	1.559	0.479	3.253	0.001
seriously	1.559	0.369	4.223	0
lovely	1.460	0.460	3.174	0.002
space	1.355	0.367	3.691	0.0002
bland	-1.307	0.471	-2.773	0.006
honey	1.262	0.431	2.928	0.003
certainly	1.256	0.503	2.495	0.013
soda	1.206	0.371	3.251	0.001
cute	1.156	0.317	3.650	0.0003

5 Conclusion

The goal of this paper was to assess how simple machine learning models trained on individual counts of words contained in Yelp reviews perform in classifying and predicting review sentiment and engagement respectively. Both models were trained using LASSO regularization with 3-fold cross-validation on different regression specifications. My model of review sentiment was fairly successful at classifying whether reviews were positive or negative, as it had a sensitivity of just under 80% and a specificity of around 84% when evaluated out of sample. On the other hand, my model of review engagement performed poorly when predicting engagement both in and out of sample. When using a post-LASSO procedure to assess the statistical significance of coefficients on variables chosen by the LASSO model, it was still unclear how any of the words included would have any impact on how frequently people engage with a given review.

One of the main limitations of this paper is that the dataset used for analysis contained no documentation, so it is unclear whether it was even representative of all Yelp reviews in the time period it was collected. However, even if I had a representative sample, the quality of my analysis would limited by the fact that I was only training machine learning models using counts of individual words. A richer analysis might incorporate common phrases or more complex measures of sentence structure in developing models to address the problems I discussed in this paper. Additionally, using web-scraping techniques, it would be possible to extract even more information from Yelp reviews than what was contained in the dataset I used, such as the names of businesses or their addresses, which would allow for studying how sentiment varies across different geographical locations or types of establishments. Regardless, the results from my model of review sentiment illustrate that even simple models can do well at detecting the sentiment of online reviews. For future research, it may be of interest to study how models trained on review data compare to existing methods of classifying strings of text.

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